

Literature answer to the finite-support Delta-point problem from arXiv:2111.14393

FA/Banach literature-status packet

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Status

The intended finite-support form of Problem 3 in Veeorg’s *Characterizations of Daugavet- and delta-points in Lipschitz-free spaces* is answered affirmatively by Abrahamsen, Aliaga, Lima, Martiny, Perreau, Prochazka, and Veeorg, *Delta-points and their implications for the geometry of Banach spaces*, arXiv:2303.00511, Theorem 6.9.

Source question

For a pointed metric space M , write $\mathcal{F}(M)$ for its Lipschitz-free space and

$$m_{xy} = \frac{\delta_x - \delta_y}{d(x, y)}$$

for a molecule. A unit vector is a Δ -point if every slice of the unit ball containing it contains vectors arbitrarily close to distance two from it.

Immediately after an example of a finite convex combination that is a Δ -point despite its displayed summands not being Δ -points, the source asks on page 20:

Source Problem 3. *Can the point instead be represented as a finite convex combination*

$$\mu = \sum_{i=1}^n \lambda_i m_{x_i y_i}, \quad \lambda_i > 0, \quad \sum_{i=1}^n \lambda_i = 1,$$

in which every molecule $m_{x_i y_i}$ is a Δ -point?

The displayed source formula literally restricts μ to $\mathcal{M}(M)$, the set of individual molecules. Read literally, that would be tautological. The surrounding paragraph concerns finitely supported molecular combinations, and the later answering paper, coauthored by Veeorg, explicitly identifies source Problem 3 as the question for finitely supported Δ -points. That later-author-confirmed formulation is the scope recorded here.

Exact literature answer

On page 34 of arXiv:2303.00511, the authors state:

Answering Theorem 6.9. *Suppose that $\mu \in S_{\mathcal{F}(M)}$ is a Δ -point with finite support. Then μ can be written as a finite convex combination of Δ -molecules in $S_{\mathcal{F}(M)}$.*

The paper explicitly labels this theorem as corresponding to Problem 3 of the source and says that it answers the question positively. Thus the intended finite-support question is completely settled.

Proof mechanism in the answering paper

The answering paper introduces auxiliary metrics b_α , $0 < \alpha < 1$, satisfying

$$(1 - \alpha)d(x, y) \leq b_\alpha(x, y) \leq d(x, y).$$

Its preceding characterization implies that a finitely supported Δ -point μ has

$$\|\mu\|_{\mathcal{F}(M, b_\alpha)} = 1 - \alpha.$$

Because μ has finite support, the normalized vector $\mu/(1 - \alpha)$ admits a finite convex decomposition into b_α -molecules. Rewriting that decomposition with the original metric and comparing norms gives

$$1 = \|\mu\|_{\mathcal{F}(M, d)} \leq \sum_i \lambda_i \frac{(1 - \alpha)d(x_i, y_i)}{b_\alpha(x_i, y_i)} \leq \sum_i \lambda_i = 1.$$

Every inequality is therefore an equality, so

$$b_\alpha(x_i, y_i) = (1 - \alpha)d(x_i, y_i)$$

for each edge in the decomposition. The paper’s metric characterization of Δ -molecules says precisely that this equality forces each $m_{x_i y_i}$ to be a Δ -point.

Scope not answered

Problem 2 of the source asks whether every slice containing an arbitrary Δ -point contains molecules of arbitrarily small metric length. Theorem 6.9 assumes finite support and does not answer this all-elements problem. The answering paper emphasizes that not every unit-sphere element of a Lipschitz-free space is even a convex sum of molecules.

References

1. T. Veeorg, *Characterizations of Daugavet- and delta-points in Lipschitz-free spaces*, *Studia Math.* **268** (2023), no. 2, 213–233; arXiv:2111.14393. Problem 3 appears on page 20 of the arXiv PDF.
2. T. A. Abrahamsen, R. J. Aliaga, V. Lima, A. Martiny, Y. Perreau, A. Prochazka, and T. Veeorg, *Delta-points and their implications for the geometry of Banach spaces*, arXiv:2303.00511. Theorem 6.9 appears on page 34 of the cached PDF.

Human-review recommendation

Mark the intended finite-support form of source Problem 3 as literature-already-answered. Retain source Problem 2 as open unless a separate later result is identified.